

A Convention-Explicit Hopf Atlas for the Brusselator

Route-A source-local certificate HCS-C311

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Abstract

For every $A > 0$ and $B \geq 0$, we prove positive-quadrant global existence for the Brusselator and give its complete node, focus, defective, and Hopf equilibrium atlas. With explicitly normalized right/left eigenvectors we derive the full complex G_{21} , distinguish three coefficient conventions, prove uniform supercriticality, and obtain leading cycle amplitude and frequency.

1 Global existence and complete linear atlas

Consider

$$\dot{x} = A - (B + 1)x + x^2y, \quad \dot{y} = Bx - x^2y, \quad A > 0, \quad B \geq 0. \quad (1)$$

At $x = 0$ the first component equals $A > 0$, and at $y = 0$ the second is $Bx \geq 0$; hence the closed nonnegative quadrant is invariant. There $(x + y)' = A - x \leq A$, so neither coordinate can blow up in finite time. Every nonnegative solution is therefore global.

Theorem 1 (All-parameter equilibrium chambers). *Equation (1) has the unique nonnegative equilibrium $E = (A, B/A)$. Put $\tau = B - 1 - A^2$. Its Jacobian is*

$$J = \begin{pmatrix} B - 1 & A^2 \\ -B & -A^2 \end{pmatrix}, \quad \text{tr } J = \tau, \quad \det J = A^2. \quad (2)$$

Within $B \geq 0$, the increasing- B atlas is

range or boundary	type
$B < (A - 1)^2$	stable node
$B = (A - 1)^2$	defective stable node
$(A - 1)^2 < B < 1 + A^2$	stable focus
$B = 1 + A^2$	Hopf pair $\pm iA$
$1 + A^2 < B < (A + 1)^2$	unstable focus
$B = (A + 1)^2$	defective unstable node
$B > (A + 1)^2$	unstable node

Empty ranges are omitted; in particular $A = 1, B = 0$ is the stable defective boundary.

Proof. At equilibrium, adding the equations gives $x = A$, then the second gives $y = B/A$. Formula (2) follows by differentiation. Its eigenvalues are $\tau/2 \pm \sqrt{\tau^2 - 4A^2}/2$. The signs of τ and of this discriminant give every row. At equality the positive off-diagonal entry A^2 prevents J from being scalar, hence the repeated eigenvalue is defective. $\square$

2 Exact normalized Hopf coefficient

Set $B = 1 + A^2$ and translate $u = x - A$, $v = y - B/A$. The nonlinear expansion is

$$\begin{aligned} \dot{u} &= A^2u + A^2v + (A + A^{-1})u^2 + 2Auv + u^2v, \\ \dot{v} &= -(A^2 + 1)u - A^2v - (A + A^{-1})u^2 - 2Auv - u^2v. \end{aligned}$$

Fix right and adjoint eigenvectors

$$q = (1, -1 + i/A)^\top, \quad p = ((1 + iA)/2, iA/2)^\top, \quad \bar{p}^\top q = 1. \quad (3)$$

If $\mathcal{B}, \mathcal{C}$ are the second and third derivative multilinear forms, use the convention

$$G_{21} = \bar{p}^T \{ \mathcal{C}(q, q, \bar{q}) - 2\mathcal{B}(q, J^{-1}\mathcal{B}(q, \bar{q})) \} \quad (4)$$

$$+ \mathcal{B}(\bar{q}, (2iAI - J)^{-1}\mathcal{B}(q, q)) \}. \quad (5)$$

Direct substitution, with no numerical fitting, gives

$$G_{21} = - \left(1 + \frac{2}{A^2} \right) - i \frac{4A^4 - 7A^2 + 4}{3A^3}, \quad \ell_1 = \frac{\text{Re } G_{21}}{2A} = - \frac{A^2 + 2}{2A^3} < 0. \quad (6)$$

Theorem 2 (Uniformly supercritical branch). *Let $\mu = B - (1 + A^2)$. The Hopf crossing is transverse with real-part derivative $1/2$. For all sufficiently small $\mu > 0$ there is one local periodic-orbit branch modulo phase; it is stable. In the coordinate of (3),*

$$r^2 = \frac{A^2}{A^2 + 2} \mu + O(\mu^2), \quad (7)$$

$$\Omega = A - \frac{4A^4 - 7A^2 + 4}{6A(A^2 + 2)} \mu + O(\mu^2). \quad (8)$$

The first harmonic of $x - A$ has amplitude $2A\sqrt{\mu/(A^2 + 2)}$ up to an $O(\mu)$ coordinate correction.

Proof. Near zero the eigenvalues are $\mu/2 \pm i\sqrt{A^2 - \mu^2/4}$, proving transversality and absence of linear frequency detuning. The physical-time normal form is $\dot{w} = (\mu/2 + iA)w + (G_{21}/2)w|w|^2 + \text{higher terms}$. Its radial cubic coefficient is $-(A^2 + 2)/(2A^2)$, whereas ℓ_1 in (6) includes the additional division by A . Solving the radial equation gives (7); its derivative is negative at the nonzero root. The imaginary part of $G_{21}/2$ gives (8), and $u = w + \bar{w} + O(|w|^2)$ gives the first harmonic. $\square$

The face $B = 0$ is included in Theorem 1. The face $A = 0$ is not: B/A is singular and $x = 0$ becomes an equilibrium line. No global uniqueness of periodic orbits is inferred from the local Hopf theorem.

Source lineage

References

- [1] I. Prigogine and R. Lefever, “Symmetry breaking instabilities in dissipative systems. II,” *J. Chem. Phys.* 48 (1968), 1695–1700. DOI: [10.1063/1.1668896](https://doi.org/10.1063/1.1668896).
- [2] Y. A. Kuznetsov, *Elements of Applied Bifurcation Theory*, 3rd ed., Springer, 2004. DOI: [10.1007/978-1-4757-3978-7](https://doi.org/10.1007/978-1-4757-3978-7).